

Donghwan Rho

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Research Interests

- Implementing Language Models Under Homomorphic Encryption
- Mathematical Reasoning in Large Language Models
- Interpretability of Language Models

Education

Ph.D.	Seoul National University , Mathematical Sciences	Mar. 2021 – Present
	• Advisor: Ernest K. Ryu • Expected Graduation: Feb. 2026	
BS	Korea University , Mathematics Education	Mar. 2015 – Feb. 2021
	• Graduated 1st in Class • (Mandatory) Military Service: May 2017 - Jan. 2019 • Teacher's Certificate for Secondary Regular Teacher of Mathematics	

Experience

Research Internship , CryptoLab	Seoul, South Korea
• Homomorphic Encryption Optimization Team	Dec. 2025 -
Workshop Coordinator , SNU-KAIST ML Theory Workshop	Gangneung, South Korea
• Assisted in planning the schedule, selecting the venue, and managing workshop programs.	Aug. 2024
Teaching Practicum	Seoul, South Korea
• Danggok high school	Apr. 2020 - May 2020
Military Service , Auxiliary Police	Seoul, South Korea
• Assisted in maintaining public order and security.	May 2017 – Jan. 2019

Publications and Preprints

*: Equal Contribution

- | | | |
|-----|---|-----------|
| [3] | Bridging LLMs and Symbolic Phonological Rules: RLVR for the Korean Word-Chain Game
Donghwan Rho
Under review
Paper | Oct. 2025 |
| [2] | Traveling Salesman-Based Token Ordering Improves Stability in Homomorphically Encrypted Language Models
Donghwan Rho* , Sieun Seo*, Hyewon Sung, Chohong Min, Ernest K. Ryu
Under review
Paper | Oct. 2025 |
| [1] | Encryption-Friendly LLM Architecture
Donghwan Rho* , Taeseong Kim*, Minje Park, Jung Woo Kim, Hyunsik Chae, Ernest K. Ryu, Jung Hee Cheon | Oct. 2024 |

Talks & Presentations

Symposium of the Research Institute of Mathematics, Seoul National University, Talk	Nov. 2025
Title: Implementing Efficient Language Models under Homomorphic Encryption Site link, Blog post (Thanks to Yossi Frenkel), Slides	
Korean Mathematical Society Annual Meeting, Talk	Oct. 2025
Title: Homomorphically encrypted language models stabilized with TSP-sorted tokens Link	
SNU-Samsung Electronics Collaboration Symposium, Poster Presentation	Aug. 2024
Title: Encryption-Friendly LLM Architecture	
SNU-KAIST ML Theory Workshop, Poster Presentation	Aug. 2024
Title: Encryption-Friendly LLM Architecture	

Honors and Awards

Outstanding TA, Seoul National University Link Introduction to Computer Programming and Artificial Intelligence for Scientists	July 2025
University Students Contest of Mathemaitcs, Silver Prize Link	Dec. 2018
Academic Excellence Scholarship, Korea University	Aug. 2015

Teaching

Peer Tutoring in Basic Sciences, Lead Teaching Assistant	Mar. 2024 - Present
- Represented section TAs to ensure the smooth operation of the peer tutoring program. - Oversaw tutorings in Mathematics, Physics, Chemistry, Biology, and Statistics.	
Basic Calculus, Teaching Assistant	Sep. 2021 - Present
- Matched tutors and tutees for the <i>Basic Calculus</i> peer tutoring program. - Managed program logistics to ensure efficient operation.	
SNU Pre-College Program, Head TA, Instructor	
- Managed logistics of the mathematics program for freshman undergraduate. - Gave lectures. - Winter 2024, Winter 2023, Winter 2022	
Introduction to Computer Programming and Artificial Intelligence for Scientists, Instructor	Spring 2025
- English Course - Selected as an outstanding TA	
Topics in Machine Intelligence	Fall 2024
- Topics: Continuous-depth neural networks, Diffusion Models, Large Language Models, Vision Language Models, State-space Models - English Course	

- Graduate Course

Mathematical algorithms II

Fall 2022

- Topics: Reinforcement Learning, Diffusion Models, Natural Language Processing
- English Course
- Graduate Course

Calculus 2 Practice, Instructor

- Fall 2021, Fall 2025

Calculus 1 Practice, Instructor

- Spring 2024, Fall 2023, Spring 2023, Spring 2022, Spring 2021

Technical Skills

Languages: Korean (Native), English (Fluent)

Programming Languages: Python, C++

Tools & Technologies: LaTeX, GitHub